ONLINE MASTERS IN **DATA SCIENCE**

DSC 257R - UNSUPERVISED LEARNING

THE GIBBS SAMPLER

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Gibbs Sampler

Finite state space $\mathcal{X} = \mathcal{X}_o^N$. Want to sample from a distribution $P > 0$ on $\mathcal{X}$:

- Start at any $x \in \mathcal{X}$
- Repeat:
  - Pick a coordinate $i \in \{1,2,\ldots,N\}$ at random
  - Resample x_i from $P(X_i = x_i | x_{\setminus \mathrm{i}})$

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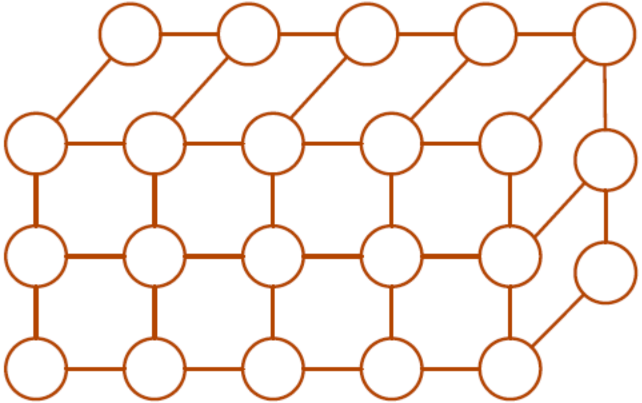
1. **There is a unique stationary distribution.**
   There is some number of steps k such that the walk can move from any location in $\mathcal{X}$ to any other location within k steps.

2. **The stationary distribution is** P.
   Check: $P(x)\mathrm{Pr}(\text{move } x \longrightarrow x') = P(x')\mathrm{Pr}(\text{move } x' \longrightarrow x)$ for any $x, x' \in \mathcal{X}$.

Example: Ising Model

System of N particles arranged in a lattice.

- Each particle has a **spin** $X_i \in \{-1, +1\}$
- Overall configuration $X = (X_1, \dots, X_N) \in \{-1, +1\}^N$



Probability $P(x) \propto e^{-U(x)}$

Energy of configuration x:

$$U(x) = -\sum_{i,j \text{ neighbors}} J_{ij} x_i x_j - \sum_i \beta_i x_i$$

Gibbs Sampler for Ising Model

Pick a coordinate $k \in [N]$ and resample its value $X_k \in \{-1, +1\}$ while keeping everything else fixed. Recall $P(x) \propto e^{-U(x)}$ for

$$U(x) = -\sum_{i,j \text{ neighbors}} J_{ij} x_i x_j - \sum_i \beta_i x_i$$